

Jash Shah

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Education

Carnegie Mellon University

08/2022 – 06/2024 • Pittsburgh, PA

Master of Science in Robotic Systems Development • GPA: 3.92/4.0

• **Key Courses:** Learning for 3D Vision, Computer Vision, Visual Learning and Recognition, Robot Autonomy, AI Venture Studio

Manipal Institute of Technology

06/2017 – 05/2021 • Manipal, India

B.Tech Mechatronics (Minor in Robotics) • GPA: 8.78/10

Experience

Robotics Researcher

05/2023 – Present • Santa Clara, CA

Nissan Advanced Technology Center (Internship: May-Aug 2023, Full-time from Aug 2024)

- Engineered graph-based transformer pipeline for multi-object tracking, achieving 0.71 AMOTA with real-time inference on vehicle
- Developed Sparse4Dv3 LiDAR-Camera fusion tracker on nuScenes; hit 0.650 AMOTA, +0.17 over camera-only baseline
- Optimized 2b-param ViT for simulation; reached 9.2% MFU, 29x speedup by resolving silent upcasting and bottlenecks via profiling
- Built remote driving system's online pose estimation module using C++ and ROS achieving 50cm lateral and 0.5 deg yaw accuracy
- Pioneered first end-to-end LiDAR+Camera tracker (raw sensors to tracks, no separate detector); competitive with two-stage SOTA
- Engineered RANSAC-based ground plane detection with CUDA acceleration, achieving 10Hz real-time processing of point cloud
- Fused LiDAR data with on-board odometry using lag compensated EKF, resulting in ~43% improvement in vehicle pose estimate
- Designed Hungarian track-matching architecture to identify ego vehicle from LiDAR tracked objects with 96% accuracy in real-time

Machine Learning Satellite Researcher

08/2023 - 08/2024 • Pittsburgh, PA

Robotics Exploration Lab, CMU

- Led a team of 5 to design a 2-stage ML approach for region classification and landmark detection to run on low-power edge devices
- Implemented an SPP classifier in PyTorch for vision-based region classification to run on NVIDIA Jetson Nano, achieving 94% mAP
- Augmented Landsat-8/9 data with conditional GANs (cGANs), simulating cloud cover and diurnal variations for dataset diversity
- Fine-tuned 18 YOLOv8n models as lightweight landmark detectors, reducing localization error to under 50 km at 700 km altitude

Computer Vision and Deep Learning Intern

01/2021 – 05/2021 • Hyderabad, India

RapidAI Vision, Defence Research and Development Organisation (DRDO)

- Architected vision-based precision landing system for a swarm of UAVs using fiducial marker landing pads from a height of 9 meters
- Conducted SIL, HIL tests with PX4, MAVROS and Gazebo, achieving 4% landing pad width error using embedded marker protocol
- Deployed custom ORBSLAM2 variant to reconstruct site with 90% accuracy using monocular camera and localized UAV in real-time

Projects

Open Source: Reviewer at huggingface/lerobot; 36 merged PRs across 12 repos: gtsam, lerobot, jax, pytorch/rl, rerun, transformers, accelerate

gtsam-splatfactors: Gaussian Splatting meets Factor Graph SLAM

06/2026 – 08/2026

Featured on the official GTSAM blog

- Expressed gsplat rendering as GTSAM factors (photometric residuals with se(3) Jacobians inside iSAM2), adding loop closure, IMU fusion, and pose covariance that gradient-descent 3DGS-SLAM pipelines lack; up to 85% ATE reduction on KITTI

lerobot-doctor: Dataset Quality Diagnostics for LeRobot

05/2026 – Present

Live in official HuggingFace LeRobot tooling

- Built a 12-check dataset diagnostic tool (timestamps, actions, video integrity, real URDF joint limits); shipped on PyPI and integrated as the Doctor tab of HuggingFace's official LeRobot Dataset Visualizer

Zero-Shot Semantic Mapping for Mobile Manipulation

04/2026

RoboHacks @ Y Combinator

- Built spatial memory layer fusing Gemma VLM embeddings with SLAM pose graph for zero-shot object retrieval in novel scenes
- Trained ACT policy on Innate OS for 6-DoF pick-and-place; integrated semantic map for language-conditioned task dispatch
- Won Innate Grand Prize + Best Use of Data (YC, 120 builders); shipped full stack on MARS robot in under 2 days

Automated Remote Driving using External Perception Sensors

08/2022 – 08/2024 • Pittsburgh, PA

Nissan North America (Capstone Project)

- Explored a novel I2V stack to estimate yaw and lateral deviation from BEV, enabling 75% lane-keeping accuracy at 22.5 cm/s
- Built bundle adjustment pipeline to refine multi-camera calibration using geometric constraints and fiducial correspondences

Skills

Programming Languages: Python, C++, C, Java, JavaScript, MATLAB, Simulink, Bash

Frameworks and Libraries: OpenCV, PyTorch, CUDA, ROS, TensorFlow, Point Cloud Library, Ardupilot, PX4 Autopilot

Tools and Platforms: Docker, Gazebo, AirSim, Git, GitLab, AWS, GCP, Azure, Apache Kafka